Chiara Camerota

Data Science · Cyber-Physical and Communication-Constrained Systems

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Summary

Postdoctoral researcher at Saint Louis University developing efficient machine learning for resource-constrained edge and IoT systems, with a statistical foundation (MSc in Statistics, Actuarial and Financial Science; PhD in Computer Science). My work spans distribution-free minimum sample-size bounds for ML classification and robustness to synthetic training data, federated learning for variance reduction and network throughput, CNN-based electrical fault and anomaly detection from thermal imagery and sensor networks, and runtime-adaptive models deployed on NVIDIA Jetson. I apply these to industrial automation, predictive maintenance, and smart agriculture, including testing models on a customized rover for pest detection.

Education

PhD in Information Engineering, DaCoNets

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University of Florence — *Nov. 2021 – June 2025*

Dissertation's title: **Machine Learning application to IoT and IIoT security and reliability**
Advisors: Tommaso Pecorella, Andrew D. Bagdanov

Bachelor's in Statistics and Master in Statistics and Financial Science

101/110 - 110/110 Summa cum laude

University of Florence — *Sept. 2013 – Feb. 2019*

Dissertation's title: **Graphical Models for circular data**
Advisors: Agnese Panzera, Anna Gottard

Research Experience

Postdoctoral researcher

St. Louis, MO

Saint Louis university — *Jan. 2025 - Ongoing*

Advisor: Flavio Esposito
Image classification for malware detection in a communication-constrained system – Smart Farm systems – AI model compression for medical purposes – AI model compression for IoT devices

Visiting student researcher

St. Louis, MO

Saint Louis University — *Jan. 2024 - Jul. 2024*

Resilience analysis of machine learning models applied to synthetic images

Researcher

Florence, Italy

DISIT Lab, University of Florence — *Apr. 2021 - Nov. 2021*

Study and development of IOT platforms and solutions to process data in the framework of smart city and industry 4.0 – Web Developer for Snap4city.org

Teaching Experience

Guest lectures for Web Technologies, Computer Networks, and AI Capstone courses

St. Louis, MO

Department of Computer Science, Saint Louis University — *Fall 2025*

TEACHER Lecture on DNS over HTTPS (DoH)
TEACHER Lecture on CSS
TEACHER How to implement a video game with only an HTML file

Guest lectures for Web Technologies, Computer Networks, and AI Capstone courses

St. Louis, MO

Department of Computer Science, Saint Louis University — *Fall 2025*

TEACHER Lecture on Chameleon Cloud
TEACHER Lecture on Transformer-based architecture
TEACHER Lecture on Docker

Guest lecture for IoT and Smart Cities courses

Florence, Italy

School of Engineering, University of Florence — *Fall 2021*

TEACHER Lecture on IoT platform
TEACHER Lecture on graph model for ontology

Teaching Assistant for Statistics course — *Florence, Italy*

School of Economics, University of Florence — *Fall 2017 - Fall 2018 AND Fall 2019*

Lectures planning

Service, Honors & Awards

2026	**Winner**, Recognized for outstanding reviewing contributions for the International Conference on Machine Learning (Golden Review)	*ICML*
2026	**Recipient**, Awarded academic research support through NVIDIA's grant program	*NVIDIA*
2025	**Winner**, SIGCOMM travel grant	*Coimbra, Portugal*
2024	**Winner**, International Association for Pattern Recognition Research Scholarship Program	*Florence, Italy*
2023	**Winner**, International B5G Student Workshop	*Madrid, Spain*

2026 **Reviewer**, IEEE International Conference on Computer Communications (INFOCOM)
2026 **Reviewer**, Annual AAAI Conference on Artificial Intelligence (AAAI)
2026 **Reviewer**, Neural Information Processing Systems (NeurIPS)
2026 **Reviewer**, International Conference on Machine Learning (ICML)

Publications

[1] Guo, L., Camerota, C., Mahmoud, M. & Esposito, F. Expert Branches: Module Diversity for Stronger Feature Learning in Laparoscopic Segmentation. *Medical Imaging With Deep Learning*. (2026)

[2] G. Donati, C. Camerota, M. Mugnaini, M. Basso, and J. T. Sawicki, "FRF-based Crack Localization in AMB-Supported Rotors Using Neural Networks," Submitted to *Mechanical Systems and Signal Processing*, MSSP25-6225, 2026.

[3] Tsutsumi A, Camerota C, Ali W, Esposito F, Madsen EC, Herman RS, Reddy P, Qiu J, Huang R, Taylor MB, Ikeyama T. Machine Learning Prediction of Pediatric In-Hospital Survival Before Extracorporeal Membrane Oxygenation Cannulation. *ASAIO Journal*. 2026; Published online July 3. doi:10.1097/MAT.0000000000002785.

[4] J. Palomares, C. Camerota, E. Coronado, C. Cervelló-Pastor, M. Shualb Siddique, and F. Esposito, "Taming Bandwidth Bottlenecks in Federated Learning via ECN-based Gradient Compression," in *Proc. 21st International Conference on Network and Service Management (CNSM)*, 2025.

[5] L. Becchi, C. Camerota, M. Intravaia, M. Bindi, A. Luchetta, and T. Pecorella, "Load Profile Generation for Robust Optimization: A Stochastic Approach Based on Conditional Probability Approximation," in *Proceedings of the 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I and CPS Europe 2025)*, pp. 1–6, 2025.

[6] C. Camerota, M. Babaiasl, N. Shakoor, and F. Esposito, "Early Detection and Management of Crop Pests Using Quadruped Robotic Systems: Challenges and Open Problems," in *IEEE C4CPS Workshop on Converged Communication, Computing, and Control for Next-Generation CPS*, 2025.

[7] C. Camerota, T. Pecorella, and A. D. Bagdanov, "The intrinsic convenience of federated learning in malware IoT detection," in *Proc. 20th International Conference on Network and Service Management (CNSM)*, pp. 1–7, 2024.

[8] C. Camerota, L. Pappone, T. Pecorella, and F. Esposito, "Addressing Data Security in IoT: Minimum Sample Size and Denoising Diffusion Models for Improved Malware Detection," in *Proc. 20th International Conference on Network and Service Management (CNSM)*, pp. 1–7, 2024.

[9] G. Donati, M. Basso, M. Mugnaini, and C. Camerota, "A Convolutional Neural Network to Locate Unbalance in Turbomachinery Supported by AMBs," in *IEEE MetroInd4.0 & IoT*, pp. 274–279, 2024.

[10] G. Donati, M. Basso, G. A. Manduzio, M. Mugnaini, T. Pecorella, and C. Camerota, "A convolutional neural network for electrical fault recognition in active magnetic bearing systems," *Sensors*, vol. 23, no. 16, p. 7023, 2023.

[11] P. Bellini, C. Camerota, and P. Nesi, "Automating Heterogeneous IoT Device Networks from Multiple Brokers with Multiple Data Models," in *Global IoT Summit*, pp. 226–238, 2022.

Remarks